GPT-to-GPT handoff reliability: a research evidence base

LLM-to-LLM handoff reliability remains an unsolved problem with no standardized protocol, no formal success criteria, and failure rates between 41% and 87% in multi-agent systems. This matters because production AI systems increasingly depend on passing context between model instances—across sessions, providers, or agent roles—yet the evidence base reveals compound reliability decay, systematic context loss, and fabrication rates that escalate with context length. The research landscape spans peer-reviewed studies on multi-agent failure taxonomies, practitioner-documented transport failures, and emerging standards (A2A, MCP, OpenTelemetry) that address interoperability but not semantic fidelity. Critically, no published controlled experiment directly compares intra-provider versus cross-provider handoffs, and no benchmark exists for measuring information retention quality during handoffs.

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1. Context continuity collapses at session boundaries

The most rigorous evidence for multi-session context failure comes from Microsoft Research and Salesforce Research. Their May 2025 study ("LLMs Get Lost In Multi-Turn Conversation," arXiv:2505.06120) analyzed over **200,000 simulated conversations across 15 LLMs** including GPT-4.1, Gemini 2.5 Pro, Claude 3.7 Sonnet, and o3. The headline finding: **39% average performance degradation** when tasks spread across multiple turns versus single-turn presentation. (arXiv) OpenAI's o3 fell from 98.1% to 64.1% accuracy. (LogRocket) Unreliability increased by 112%. (MarkTechPost) Even two-turn conversations showed significant drops, and 78% of initial errors propagated through all subsequent turns— (Techfunnel)models that take a wrong turn do not recover. (arXiv)

The "concat-and-retry" fix (consolidating all information into a fresh prompt) restored performance to approximately 90%+, confirming that accumulated context pollution causes degradation rather than capability limitations. This directly validates the ContextKeeper hypothesis: session boundaries create lossy transitions, but structured context reconstruction can mitigate them.

Fabrication escalates with context length. The RIKER study (Roig, arXiv:2603.08274, March 2026) evaluated 35 open-weight models across 172 billion tokens in document Q&A scenarios. (arxiv) Best-performing models fabricated at a **minimum 1.19% rate at 32K context**, rising to **5–7% for top-tier models**. (arXiv) (arXiv) At 128K tokens, fabrication nearly tripled. At 200K tokens, **every model exceeded 10% fabrication**, with GLM 4.6 reaching 69.53% fabrication despite maintaining 57.94% grounding accuracy. (arXiv) This decoupling between grounding ability and fabrication resistance is particularly dangerous for handoff scenarios: a model can correctly find facts in a handoff document while simultaneously fabricating facts that don't exist in it.

Chroma's "Context Rot" report (Hong, Troynikov, and Huber, July 2025) tested 18 LLMs and found performance degrades **nonuniformly** as input length increases, even on simple tasks. trychroma All models showed significantly higher performance on focused ~300-token prompts versus full ~113K-token prompts on LongMemEval multi-session QA tasks— Trychroma confirming that more context does not mean better context reconstruction.

Multi-session memory benchmarks quantify the gap starkly. LongMemEval (Wu et al., arXiv:2410.10813, October 2024) found long-context LLMs show **30–60% performance drops** on full context versus focused context, arXiv with GPT-3.5-turbo scoring 20.3 on temporal reasoning versus 92.6 for humans. Emergent Mind MEMTRACK found that even GPT-5 achieves only **60% correctness** on cross-session information tracking. arXiv A University of Michigan/Amazon study (Chung et al., arXiv:2512.04307, December 2025) showed web agent success rates dropping from 40–50% to under 10% in long-context scenarios arXiv of 25K–150K tokens. alphaXiv

A critical practitioner-documented bug in Claude Code (GitHub Issue #29160, anthropics/claude-code) revealed that during context compaction, **LLM-generated text appeared as human (user) turns**—a trust boundary violation where reconstructed context corrupted message authorship. The reporter noted: "If the fabricated message had said 'yes, proceed' or 'delete the branch,' the model would have acted on it." GitHub

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2. Multi-agent handoff failure rates range from 41% to 87%

The most comprehensive taxonomy of multi-agent system failures comes from UC Berkeley's MAST study (Cemri et al., arXiv:2503.13657, accepted at ICLR 2025—note: also reported as NeurIPS 2025 Datasets and Benchmarks Track). Across 1,642 annotated execution traces from 7 frameworks (MetaGPT, ChatDev, HyperAgent, OpenManus, AppWorld, Magentic, AG2), the researchers identified **14 distinct failure modes** in three categories: specification/design failures, inter-agent misalignment, and task verification/termination problems. arXiv arXiv **Failure rates ranged from 41% to 86.7%** across frameworks. arXiv Coordination breakdowns accounted for **36.9% of all failures**. Towards Data Science Interventions including improved specifications and enhanced orchestration did not resolve failures, suggesting fundamental architectural issues rather than prompt-fixable problems. arXiv

Google DeepMind's scaling study (Kim et al., arXiv:2512.08296, December 2025) tested 180 agent configurations across 5 architectures and 3 LLM families. X arXiv **Unstructured multi-agent networks amplified errors by 17.2× compared to single-agent baselines.** Centralized coordination contained amplification to 4.4×. Google Research For sequential reasoning tasks, all multi-agent variants degraded performance by **39–70%**. X arXiv Capability saturation emerged as a key finding: once single-agent baselines exceeded approximately 45% accuracy, adding agents provided diminishing or negative returns X ($\beta=-0.408$, $p<0.001$). Emergent Mind

The mathematical foundation is stark. As documented by Dziri et al. (2023, "Faith and Fate"): with 90% per-step accuracy, a 10-step process achieves only **35% end-to-end reliability** (0.9^10 = 0.35). Tushardadlani At 95% per-step across 20 steps, reliability falls to 35.8%. Towards Data Science This compound decay is the fundamental constraint on multi-agent handoff chains.

Production systems respond by staying simple. The MAP study (Pan et al., arXiv:2512.04123, December 2025) surveyed 306 practitioners across 26 industry domains arXiv (Accenture, Amazon, AMD, Google, IBM, Intel, Samsung SDS, SAP). ImagineX Key findings: **68% of production agents execute at most 10 steps** before human intervention, 70% rely on prompting off-the-shelf models with no fine-tuning, and 74% depend primarily on human evaluation. arXiv Reliability was rated the number-one development challenge. Emergent Mind No team reported applying standard production reliability metrics such as "five 9s" availability to agent systems. Dakshineshwari Production agents are far simpler than academic literature suggests. Dakshineshwari

The NESTFUL benchmark (Basu et al., arXiv:2409.03797, EMNLP 2025) specifically tested chained API call reliability across 1,800+ nested sequences. **GPT-4o achieved only 28% full-sequence match accuracy**, OpenReview demonstrating that models understand individual tools but struggle fundamentally with chained composition. DEV Community Performance dropped significantly with increased nesting depth and data dependencies. The Moonlight

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3. Document reading fails predictably by position, length, and transport mechanism

The landmark "Lost in the Middle" study (Liu et al., *Transactions of the Association for Computational Linguistics* 12:157–173, 2024) established that LLM attention follows a **U-shaped curve**: highest for information at the beginning or end of context, with **20+ percentage point accuracy drops** for mid-positioned content. MIT Press ACL Anthology GPT-3.5-Turbo's accuracy on multi-document QA fell **below its closed-book baseline (56.1%)** when relevant information was placed mid-context— MIT Pressmeaning added context actively hurt the model. Teapot123 Even explicitly long-context models showed identical degradation. arXiv Key-value retrieval accuracy dropped to approximately 1% at certain mid-context positions. Maxim Articles MIT Press

A striking 2025 finding (arXiv:2510.05381) demonstrated that **context length alone hurts performance even when retrieval is perfect**. When models correctly recited every token of evidence (100% exact match), reasoning performance still degraded substantially with input length. arXiv Inserting 25,000 white spaces—minimal distraction—caused models to reach wrong answers despite correctly extracting all conditions. arXiv

Document grounding benchmarks reveal persistent unfaithfulness. Google DeepMind's FACTS Grounding Leaderboard (Jacovi et al., arXiv:2501.03200, January 2025) evaluated 1,719 examples Google DeepMind across finance, technology, medicine, and law with documents up to 32K tokens. Top-performing models scored **83–88% grounding accuracy**, MarkTechPost meaning

even the best models fail to fully ground **12–22% of responses** in provided documents. GPT-4o scored 78.7–83.6%; Claude 3.5 scored 83.3%. Models excelled on technical/financial tasks but struggled with medical and legal contexts. MarkTechPost

The RAGTruth corpus (Niu et al., ACL 2024) provided 18,000 manually annotated RAG responses arXiv and found hallucinations persist even with retrieved context. GPT-4-turbo achieved only **68.3% F1** in detecting its own hallucinations. arXiv A particularly concerning finding from Wallat et al. (ACM SIGIR ICTIR 2025) showed that **up to 57% of RAG citations were post-rationalized**—models generated answers from parametric memory then superficially searched for supporting documents. ResearchGate +3 When adversarial statements were injected into random documents, models cited them 12% of the time. ResearchGate arXiv

Prompt injection through documents is nearly universal. Greshake et al. (ACM AISec 2023) demonstrated that LLM-integrated applications blur the line between data and instructions. Evidently AI A medical domain study (PMC, 2025) found prompt-injection attacks succeeded in **94.4% of 216 trials** and 91.7% of extremely high-harm scenarios including Category X pregnancy drugs. This represents universal susceptibility across flagship models and directly threatens document-mediated handoff reliability.

Specific GPT-4 document reading accuracy was measured by Osman et al. (*BMC Medical Research Methodology* 24:139, 2024) on 3,600 queries against 60 HIV drug resistance papers. **Mean accuracy was 86.9%** with 72.5% recall and 87.4% precision. PubMed Instruction sheets with domain knowledge did not significantly improve accuracy. Significant performance drops occurred for lengthy documents. arXiv

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4. Three transport mechanisms fail in distinct, documented ways

Project-loaded files exhibit the most dangerous failure mode: silent hallucination. Multiple OpenAI Community Forum reports (2024–2025) document ChatGPT Projects reading only the first approximately 2,000 words of project files and hallucinating the remainder. OpenAI Developer Commu… Files over 12,000 words triggered fabrication; files under 12,000 words were read but content beyond approximately 10,000 words was simply absent. openai Users documented ChatGPT fabricating responses using file title information plus training data rather than actual file contents—responses that appeared grounded but were entirely confabulated. OpenAI Developer Commu… An April 2025 GitHub issue (openai/openai-cookbook #1780) documented ephemeral sandbox loss where files appear in the sidebar but become inaccessible mid-session after environment reset, with no user notification. GitHub

Claude Projects has distinct failure patterns. A detailed GitHub bug report (anthropics/claude-code #25759) documented that Claude switches from direct context loading to RAG search mode at approximately **13 files regardless of total token size**. A project with 15 files totaling only 73,000 tokens (35% of the 200K window) was forced into RAG mode, contradicting Anthropic's

documentation. GitHub A persistent caching bug (issue #10841, November 2024) showed that when files are deleted, modified, and re-uploaded—even with new filenames—Claude continues referencing old content. The cache appears content-indexed rather than filename-based, creating version drift that no user action could resolve. GitHub

Prompt-box attachments suffer primarily from positional bias (the Lost-in-the-Middle effect, 30%+ degradation for mid-positioned information) and token overflow. Maxim Articles ChatGPT enforces hard limits: 512MB per file, 2M tokens per file, 10 files per batch, approximately 128K context window. OpenAI Help Center When content exceeds context length, the system returns a hard error causing complete conversation reset with loss of all prior context. GitHub ChatGPT does not load entire files into context but builds a "private semantic index" and retrieves chunks per query—making file handling inherently RAG-based, not full-context, even when users believe otherwise. Data Studios

API/connector-based retrieval (RAG) fails through embedding drift—a silent degradation where the same text produces different vectors over time due to model updates, preprocessing changes, or partial re-embedding. Decompressed Documented cases show recall dropping from **0.92 to 0.74** with no log entries explaining why. Decompressed A practitioner catalog (Paul, DEV Community, October 2025) identified 10 systematic RAG failure modes including retrieval timing attacks (generation beginning before retrieval completes), cross-document contradictions (no built-in resolution mechanism), temporal staleness, and citation hallucination. dev DEV Community An Elasticsearch Labs comparison (Rengifo and Martin, July 2025) found that full-context processing of 1M tokens was significantly slower and more expensive than RAG, while producing more inaccuracies when multiple articles contained similar information. Elastic

Documented API-versus-chat-interface performance gaps compound transport problems. Developers on the OpenAI Community Forum (February 2024) reported that identical GPT-4 models via API produced significantly worse PDF results than the ChatGPT UI on identical documents. openai **No single peer-reviewed comparative study systematically evaluates all three transport mechanisms** (project files, prompt attachments, RAG) against each other for context delivery reliability. This is a significant gap in the literature.

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5. LLM output is non-deterministic even at temperature zero

The most rigorous study of LLM non-determinism comes from Atil et al. ACL Anthology (ACL Eval4NLP Workshop 2025, peer-reviewed). Testing 5 API-based LLMs across 8 tasks with **10 identical runs at temperature=0**, they found arXiv **accuracy variations of up to 15%** and a gap of up to **70% between best and worst possible performance** aggregated across runs. ACL Anthology aclanthology Accuracy distributions were non-normal (Kolmogorov-Smirnov test rejected normality with $p < 10^{-9}$). aclanthology Locally run LLMs demonstrated stability,

confirming the issue resides in hosted/API environments. aclanthology The study introduced novel metrics: TARr@N (Total Agreement Rate for raw output across N runs) and TARa@N (for parsed answers), providing standardized measurement of non-determinism. aclanthology

The technical root cause is now well understood. Sherman Chann's investigation (August 2023) and a detailed Thinking Machines Lab analysis (September 2025, Horace He) identified **Sparse Mixture-of-Experts routing** as the primary cause. Batched inference with MoE means tokens from different sequences compete for expert capacity, producing sequence-level non-determinism even when individual GPU kernels are deterministic. 152334H thinkingmachines A Towards Data Science investigation identified **12 clusters of log probabilities** in GPT-4o outputs, potentially corresponding to 12 different MoE experts—with log probabilities themselves being non-deterministic regardless of temperature and seed settings. Towards Data Science

Ouyang et al. (*ACM Transactions on Software Engineering and Methodology* 34(2), 2025) tested GPT-3.5 and GPT-4 on code generation benchmarks. At default settings, **more than 50% of problems** had not a single equal test output among top-5 candidates. Maximum test pass rate difference reached 100% for all three datasets. arXiv Even at temperature=0, 18–44% of problems produced no equal test output across 5 candidates. arXiv

No major provider guarantees deterministic outputs. OpenAI's documentation states results are "mostly deterministic"; Anthropic states "results will not be fully deterministic"; Google provides similar disclaimers. Unstract The practical consensus is to design systems robust to minor variations rather than expecting bit-perfect reproducibility. Unstract

The evaluation framework landscape is mature but fragmented. IFEval (Zhou et al., Google, arXiv:2311.07911, November 2023) provides approximately 500 prompts with programmatically verifiable instructions— arXivno LLM-based judging needed. MT-Bench (Zheng et al., NeurIPS 2023) offers 80 multi-turn questions with validated LLM-as-judge achieving greater than 80% agreement with human preferences. arXiv OpenReview HELM (Liang et al., Stanford CRFM, *Annals of the New York Academy of Sciences*, 2023) evaluates across 7 metrics and 42 scenarios. arXiv OpenReview EleutherAI's lm-eval harness provides task versioning, YAML-based configs, and caching for reproducible evaluation, EleutherAI supporting 200+ tasks and used internally by NVIDIA, Cohere, and others. The Biderman et al. companion paper (arXiv:2405.14782, May 2024) documents three years of experience and notes that teams "frequently struggled to reproduce results reported in various papers." arXiv

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6. Append-only provenance is converging on OpenTelemetry and W3C PROV

LLM workflow tracking differs fundamentally from classical ML. Classical ML tracks training runs (parameters, metrics, model artifacts). LLM tracking must capture individual inference requests with full prompt/response content, token usage, cost, latency, MLflow multi-step chain execution, and tool calls—creating orders of magnitude more log volume. MLflow's GenAI-

focused tracing (mlflow.org) captures inputs, outputs, and metadata at each intermediate step within a single request, with auto-tracing for 15+ libraries. MLflow Weights & Biases Weave tracks tool call monitoring, handoff tracing, and state evolution for multi-agent systems. Both are fully OpenTelemetry-compatible. MLflow

OpenTelemetry is emerging as the de facto standard for LLM observability. The GenAI Semantic Conventions SIG (started April 2024) defines standard attributes (`gen_ai.operation.name`, `gen_ai.request.model`, `gen_ai.usage.input_tokens`, etc.) with provider-specific conventions for Anthropic, Azure, AWS Bedrock, and OpenAI. OpenTelemetry New proposed conventions for agent systems cover Tasks, Actions, Agents, Teams, Artifacts, and Memory. Nearly every major tool—MLflow, Langfuse (21,000+ GitHub stars, MIT license), LangSmith, Helicone (processed 2+ billion LLM interactions), Helicone Datadog, and New Relic—now supports or builds on OpenTelemetry.

The W3C PROV data model has been explicitly extended for AI agent workflows. **PROV-AGENT** (Souza et al., Proceedings of the 21st IEEE International Conference on e-Science, 2025; arXiv:2508.02866), from Oak Ridge and Argonne National Laboratories, extends W3C PROV with Model Context Protocol integration. arXiv arXiv It models AIAgent as a subclass of W3C PROV Agent, captures prompts, responses, decisions, and downstream impacts, and includes scheduling and telemetry data as PROV Entity subclasses. arXiv DLProv (Pina et al., *PeerJ Computer Science* 11:e2985, July 2025) provides a W3C PROV-compliant suite with maximum overhead of only **1.4%** in execution time. PubMed Central ResearchGate MLflow2PROV (Schlegel and Sattler, ACM SIGMOD DEEM'23 Workshop; *Information Systems* 132:102495, July 2025) extracts W3C PROV-compliant provenance graphs from existing MLflow instances with no code changes required. ScienceDirect

Regulatory pressure creates binding requirements. The EU AI Act Articles 12 and 19 mandate automatic logging, minimum 6-month retention, and full traceability for high-risk AI systems. EU Artificial Intelligence Act Three of Europe's most impactful AI enforcement cases in 2024 were triggered by logging failures. ISMS.online Financial institutions may require 7+ years of retention. ISO 42001 aligns with these requirements, treating AI logs as legal instruments enforceable by audit, investigation, or court. ISMS.online The AI reproducibility crisis underscores the urgency: only approximately 5% of AI researchers share source code, AIMultiple and nearly 70% cannot reproduce others' results (Semmelrock et al., *AI Magazine*, 2025; Desai et al., *AI Magazine*, 2025).

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7. No validated handoff protocol exists, but emerging standards are converging

The central finding for the ContextKeeper whitepaper is a confirmed research gap: no published controlled experiment directly compares handoff reliability across transport mechanisms, across providers, or between intra-provider and cross-provider handoffs. No standardized state serialization protocol exists for LLM handoffs. No formal specification defines

what constitutes a "successful" handoff beyond downstream task completion. No benchmark measures information retention quality during handoffs.

The closest approaches to handoff protocols in production are framework-specific:

- **OpenAI Swarm/Agents SDK** implements function-based handoffs where returning an Agent object transfers control, Openai with full conversation history carried forward. It is explicitly stateless with no persistent state between calls. galileo GitHub The Agents SDK adds Sessions for persistent memory OpenAI but remains vendor-specific.
- **LangGraph** uses graph-based state machines with typed schemas and durable checkpoints enabling pause/resume. Towards Data Science It processes 67% fewer tokens than the handoff pattern through context isolation in a subagents pattern. Langchain Used by Klarna in production (2.3 million conversations/month, resolution time reduced from 11 minutes to under 2 minutes). Towards Data Science LangChain
- **Microsoft Agent Framework (AutoGen + Semantic Kernel)** implements event-driven pub-sub communication with topic-based routing. GitHub It provides `save_state()` and `load_state()` methods for serializing full team state. Documented GitHub issues show agents getting stuck in loops instead of handing off, GitHub and agents assuming other agents' tool calls are available. GitHub
- **Anthropic's multi-agent research system** (June 2025 engineering blog) uses filesystem/artifact handoffs where subagents write to external storage and pass lightweight references back, explicitly to minimize "game of telephone" effects. Anthropic Multi-agent systems use **15× more tokens** than single-agent chat; token usage alone explains 80% of performance variance. Anthropic anthropic

Google's Agent2Agent (A2A) protocol (launched April 2025, donated to Linux Foundation June 2025, 150+ supporting organizations by v0.3) Apono is the most significant emerging standard. Built on JSON-RPC 2.0 over HTTPS Google Developers with Agent Cards for capability discovery, Google Apono it explicitly supports agents that "don't share memory, tools and context." Google Developers A2a-protocol It complements Anthropic's MCP (agent-to-tool connections) rather than replacing it. Cloudsummit +2 However, A2A addresses interoperability and routing—not semantic fidelity of context transfer.

Cross-provider handoff challenges are documented but not systematically studied. VentureBeat's analysis (2025) identified tokenizer incompatibilities (Anthropic's tokenizer breaks text into more tokens than OpenAI's), output format biases (GPT-4o biases toward JSON; Anthropic models equally adhere to JSON or XML), and differing context window performance profiles (GPT-4 most performant up to 32K tokens; Sonnet-3.5 declining beyond 8K–16K despite a 200K window). VentureBeat No academic work compares these differences in controlled handoff experiments.

Context compression research offers partial solutions. Acon (arXiv:2510.00615, October 2025) reduces memory usage by **26–54% while preserving over 95% of accuracy**. (arXiv) Practitioner approaches (Claude Code's `/compact` command, (GitHub) Factory.ai's rolling summaries) use structured JSON summaries with fields like session_intent, progress_so_far, key_facts, artifacts_created, and next_steps. (GitHub) General context compression claims **50–80% token reduction** while preserving needed information. (OneUptime) However, these approaches are ad hoc—each framework implements its own compaction strategy with no benchmark for measuring retention quality.

Conclusion: seven evidence-backed design imperatives

The research evidence converges on several findings directly relevant to ContextKeeper's experimental design. First, **session boundaries are lossy by default**—the Microsoft/Salesforce 39% degradation finding (arXiv) and the RIKER fabrication escalation data confirm that naive context transfer fails predictably. Second, **compound reliability decay is the fundamental mathematical constraint**—any per-step error rate, no matter how small, causes failure in sufficiently long sequential processes. (Preprints.org) Third, **transport mechanism matters but no comparative study exists**—ChatGPT Projects silently truncates and hallucinates, (openai) Claude Projects triggers premature RAG at 13 files, (GitHub) and RAG systems suffer embedding drift, making this comparison a genuine contribution opportunity. Fourth, **structured handoff contracts outperform free-text transfers**—hierarchical architectures with explicit validation show only 5.5% performance drops versus 23.7% for simple chains. (OpenReview) Fifth, **non-determinism is architectural, not parametric**—temperature=0 does not guarantee reproducibility (Bartosz Mikulski) due to MoE routing, requiring statistical evaluation rather than deterministic assertions. Sixth, **W3C PROV and OpenTelemetry provide mature provenance infrastructure** now being extended specifically for agent workflows. Seventh, **the intra-provider versus cross-provider handoff comparison is an unoccupied research niche**—no published work addresses this directly, positioning controlled experiments here as novel contributions to the field.

Gartner's prediction that over 40% of agentic AI projects will be canceled by 2027 (Maxim Articles +2) underscores the timeliness of this work. (Towards Data Science) The gap between academic multi-agent research and production reality—where 68% of deployed agents execute 10 steps or fewer with heavy human oversight— (arXiv)suggests that reliable handoff protocols are not just a technical convenience but a prerequisite for scaling AI agent systems beyond their current constrained deployments. (arXiv)